

Problem 1

Problem 1

Problem 1 Solve the following word problems:

- (a) The velocity function for an object moving along a line east/west is given by $v(t) = -t^2 + 4t - 3$ feet per minute.

- (i) Find the total displacement the object traveled from 2 minutes to 6 minutes (assume east is positive).

Explanation. The object's total displacement is given by $\int_2^6 v(t) dt$.

So we compute:

$$\begin{aligned}\int_2^6 v(t) dt &= \int_2^6 (-t^2 + 4t - 3) dt \\ &= \left[-\frac{1}{3}t^3 + 2t^2 - 3t \right]_2^6 \\ &= (-72 + 72 - 18) - \left(-\frac{8}{3} + 8 - 6 \right) \\ &= \frac{8}{3} - 20 = -\frac{52}{3}.\end{aligned}$$

So the object's displacement is $\frac{52}{3}$ feet west of its original location.

- (ii) Find the total distance the object traveled from 2 minutes to 6 minutes.

Explanation. First notice that $v(t) = -(t^2 - 4t + 3) = -(t-1)(t-3)$. So we can see that

$$v(t) > 0 \text{ when } 2 \leq t < 3.$$

$$v(t) < 0 \text{ when } 3 < t \leq 6.$$

Thus, the total distance that the object traveled from 2 minutes to

6 minutes is:

$$\begin{aligned}
 \int_2^6 |v(t)| \, dt &= \int_2^3 |v(t)| \, dt + \int_3^6 |v(t)| \, dt \\
 &= \int_2^3 v(t) \, dt + \int_3^6 -v(t) \, dt \\
 &= \int_2^3 (-t^2 + 4t - 3) \, dt - \int_3^6 (-t^2 + 4t - 3) \, dt \\
 &= \left[-\frac{1}{3}t^3 + 2t^2 - 3t \right]_2^3 - \left[-\frac{1}{3}t^3 + 2t^2 - 3t \right]_3^6 \\
 &= \left((-9 + 18 - 9) - \left(-\frac{8}{3} + 8 - 6\right) \right) - \\
 &\quad ((-72 + 72 - 18) - (-9 + 18 - 9)) \\
 &= \left(0 - 2 + \frac{8}{3} \right) - (-18 - 0) \\
 &= 16 + \frac{8}{3} = \frac{56}{3}.
 \end{aligned}$$

So, the total distance that the object traveled is $\frac{56}{3}$ feet.

- (iii) Suppose that the object's position 2 minutes into the trip is 5 feet of a placement marker. What is its position (relative to the placement marker) at 6 minutes.

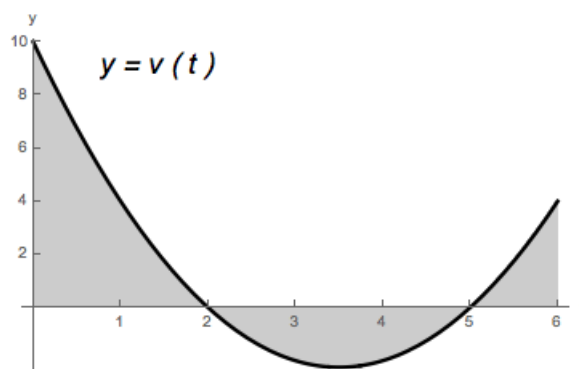
Explanation. $s(6) = s(2) + \int_2^6 v(t) \, dt = 5 + \left(-\frac{52}{3}\right) = -\frac{37}{3}.$

So the object's position at 6 minutes is $\frac{37}{3}$ feet west of the placement marker.

- (b) Sammy the Snail sets up camp in the median of I-70 and, starting at noon and ending at 6pm, hikes back and forth along the highway. He starts his hike at his campsite. His velocity at time t hours (after noon) is given by $v(t) = (t - 2)(t - 5)$ inches per hour. Find the total distance Sammy travelled on his hike.

Explanation. The total distance that Sammy travels is $\int_0^6 |v(t)| \, dt$. The following picture indicates where $v(t)$ is positive and negative:

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So we compute:

$$\begin{aligned}
 \int_0^6 |v(t)| \, dt &= \int_0^2 |v(t)| \, dt + \int_2^5 |v(t)| \, dt + \int_5^6 |v(t)| \, dt \\
 &= \int_0^2 v(t) \, dt - \int_2^5 v(t) \, dt + \int_5^6 v(t) \, dt \\
 &= \int_0^2 (t^2 - 7t + 10) \, dt - \int_2^5 (t^2 - 7t + 10) \, dt + \int_5^6 (t^2 - 7t + 10) \, dt \\
 &= \left[\frac{1}{3}t^3 - \frac{7}{2}t^2 + 10t \right]_0^2 - \left[\frac{1}{3}t^3 - \frac{7}{2}t^2 + 10t \right]_2^5 + \left[\frac{1}{3}t^3 - \frac{7}{2}t^2 + 10t \right]_5^6 \\
 &= \left(\left(\frac{8}{3} - 14 + 20 \right) - 0 \right) - \left(\left(\frac{125}{3} - \frac{175}{2} + 50 \right) - \left(\frac{8}{3} + 6 \right) \right) + \\
 &\quad \left((72 - 126 + 60) - \left(\frac{125}{3} - \frac{175}{2} + 50 \right) \right) \\
 &= -82 + 175 - 78 = 15.
 \end{aligned}$$

So Sammy has traveled a distance of 15 inches.